CS 736: Medical Image Computing

(Instructor : Suyash P. Awate)

End-Semester Exam: Closed Book / Notes / Everything

Roll Number: ______________________________

Name: ______________________________

When in doubt, make suitable assumptions, state them clearly, and proceed to solve the problem. All the best!

1. [25 points]

   Consider a set of N MRI images $\{R_n(\cdot)\}_{n=1}^{N}$ with associated binary segmentations of an object of interest indicated by the images $\{S_n(\cdot)\}_{n=1}^{N}$. Thus, at voxel location x, the MRI-image intensity is $R_n(x)$ and the binary label is $S_n(x)$.

   • (10 points: 6 + 4) Formulate an optimization problem to find (i) a mean template image for the MRI, say, $R(\cdot)$, and (ii) a mean template image for the segmentation, say, $S(\cdot)$. Assume affine transformations within registration. Also, describe an implementable algorithm.

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• (5 points) Suppose you want to use the mean template MRI image $R(\cdot)$ and the mean template segmentation image $S(\cdot)$ to segment a new MRI image $Q(\cdot)$. Formulate any optimization problem(s) required to be solved. Also, describe an implementable algorithm.

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first, solve the registration problem, involving optimizing the affine transformation between the mean template MRI image and the new MRI image (3 points)

then, spatially transform the template segmentation to the coordinate frame of the new MRI image (2 points)

• (10 points) Suppose you want to segment a new image $T(\cdot)$ without using the mean template, but using directly all information in the original dataset $\{(R_n(\cdot), S_n(\cdot))\}_{n=1}^{N}$. Formulate any optimization problem(s) required to be solved. Also, describe an implementable algorithm.

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first, solve the registration problems, involving optimizing the affine transformation between the every MRI image $R_n(\cdot)$ and the new MRI image $Q(\cdot)$ (3 points)

then, spatially transform each segmentation $S_n(\cdot)$ to the coordinate frame of the new MRI image (3 points)

then, average all the transformed segmentations (4 points)

2. [25 points]

   Consider a problem of reconstructing a 2D MRI image X (complex-valued intensities) from the acquired 2D data matrix Z (complex-valued components) involving i.i.d. zero-mean Gaussian noise.

   • (8 points: 4 + 4) Formulate a Bayesian image reconstruction problem, specifying (i) the likelihood model and (ii) an edge-preserving prior model.

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• (12 points: 4 + 8) In order to speedup the data acquisition process, suppose you acquire only a chosen subset of the data Z (you known which part of the data matrix is acquired and which isn't). Then, (i) reformulate the reconstruction problem and (ii) describe a step-by-step implementable algorithm for the image reconstruction, including the initialization, updates, and termination criterion.

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• (5 points: 1 + 4) Formulate a Bayesian image reconstruction problem, specifying (i) a likelihood model and (ii) a prior model based on a sparse dictionary representation on patches, where the dictionary is provided to you.

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3. [15 points]

Consider a dataset of N points in D dimensions, i.e., $\{x_n \in \mathbb{R}^D\}_{n=1}^N$. Consider $N \gg D$. Consider that the dataset is *centered* and spans a subspace of $\mathbb{R}^D$ having dimension $E < D$.

- (8 points) Prove mathematically that the Frechet mean, using the Euclidean distance metric in $\mathbb{R}^D$, must lie in the span of the data.

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one solution (there can be other arguments):

proof by contradiction.
let Frechet mean be ν
let the projection of the mean on the subspace that is the span of the data be μ
by pythagoras theorem, $d^2(x_i, \nu) = d^2(x_i, \mu) + d^2(\mu, \nu)$
thus, ν is a better solution to the Frechet mean
contradiction.

• (7 points) Prove mathematically that every *mode of variation* must be representable as a vector lying in the span of the data.

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4. [25 points]

   Consider a problem of clustering a group of M shapes of 3-dimensional objects, where the shapes are represented as pointsets of cardinality N. You know that the number of clusters is K.

   • (4 points) Define mathematically the standard Procrustes distance between shapes.

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• (6 points) For the specified task, using standard Procrustes distance to measure dissimilarity between shapes, design an objective function to optimize for the clustering task.

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simplest way is to perform k means using Procrustes distance

• (10 points: 7 + 2 + 1) Based on the objective function designed in the previous question, describe an implementable algorithm for the clustering, including the initialization for all unknown variables, and the stopping criterion. [Without the correct algorithm, the initialization strategy and the stopping criterion don't fetch any points.]

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(0) intialize class labels randomly
(1) for each class, solve for the new mean
align each pointset to each mean, and find procrustes distance
(2) find nearest mean to each pointset, and assign a class label to the pointset
(3) repeat steps 1 and 2
(4) terminate when the class labels don't change

• (5 points) If you are told that the shape dataset has some outliers and you want your clustering to be less sensitive to the presence of outliers in the dataset, then re-design the objective function based on concepts covered in class during the semester.

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in objective function, don't use (sum of) square of (procrustes) distances, as typically used in k-means, but use (sum of) (procustes) distances themselves.

5. [10 points]

A multivariate Gaussian probability density function (PDF) on D random variables can be parametrized using a mean vector and a precision matrix as:

$$P(x;\mu,\Lambda) := \eta(\Lambda)\exp(-0.5(x-\mu)^\top\Lambda((x-\mu))),$$

where $x \in \mathbb{R}^D$. Assume that the matrix Λ has a 3-band structure such that only the diagonal, subdiagonal, and superdiagonal elements are non-zero.

Is this multivariate Gaussian PDF modeling a PDF of some underlying Markov random field (MRF) ? If so, show the neighborhood system (smallest possible sets of neighbors for each variable), cliques, and potential functions. If not, prove that it violates some property of a MRF.

Multivariate Gaussians are MRFs.

The precision matrix is the inverse covariance matrix that is symmertic.

For a symmetric matrix Λ, the quadratic form $z^\top\Lambda z = \sum_{i,j}\Lambda_{ij}z_iz_j$.

In our case, $z_i = (x_i - \mu_i)$

In our case, $\Lambda_{ij} \neq 0$ iff $j = \max(i-1,0)$ or $j = i$ or $j = \min(i+1,D)$. So, the other $i-j$ pairs (x_i,x_j) don't interact.

For random variables $X_1, X_2, \cdots, X_D$, the neighborhood system is such the neighbors of node i are nodes $\max(i-1,1)$ and $\min(i+1,D)$.

Cliques are of 2 kinds: single nodes and node pairs where the nodes are neighbors.

Potential functions are of the form $\Lambda_{ij}(x_i-\mu_i)(x_j-\mu_j)$.